- The number of clusters will be fixed, and equal to `2`;
- The maximum number of points per cluster is bound to `M`.

We will talk diffusely about clustering and k-means in chapter 12, so please refer to section 12.2 to see the details of its implementation.

The running time for k-means, with at most `j` iterations on a dataset with `n` points, is `O(jkn)`[22], where of course k is the number of centroids.

Since for the split heuristic we have `k==2`, and the number of points in the node to split is `M+1`, the running time becomes `O(jdM)`, compared to `O(dM)` we had for the original split heuristic described in section 10.3. We can therefore trade-off the quality of the result for performance by controlling the maximum number of iterations `j`; figure 10.27 shows an example of how impactful this heuristic can be, and why the increase in running time is well worth.

